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GitHub Link	https://github.com/chaitramanohar/MLA0402-DEEP-LEARNING

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning
SLOT :

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10%
Assessment Tool Weightage: 40%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I T.Chaitra(192425217) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:T.Chaitra

Name of the student:T.Chaitra

Criteria	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

QUESTIONS: 05

Total Marks:25

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding ; some requirements unclear	Poor understanding ; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Q1. Transfer Learning for Skin Cancer Detection

Scenario

A healthcare startup has only a limited number of labeled dermoscopic images. Training a CNN from scratch may lead to **overfitting**. Therefore, transfer learning using a pre-trained CNN is suitable.

Proposed Model

Use a **pre-trained ResNet50** model.

Why ResNet50?

ResNet50 is suitable because:

- It has already learned useful visual features from a large dataset.
- Residual connections help train deep networks.
- It reduces the need for a large labeled medical dataset.
- It provides strong feature extraction capability.
- It can be fine-tuned for dermoscopic images.

Architecture

Dermoscopic Image



Resize + Normalization



Pre-trained ResNet50



Frozen Convolutional Layers



Global Average Pooling



Dense Layer + ReLU



Dropout



Output Layer + Sigmoid



Benign / Malignant

Which layers are frozen?

Initially, freeze the **early and middle convolutional layers**.

These layers learn general features such as:

- Edges
- Shapes
- Textures
- Color patterns

These features are useful for many types of images.

Which layers are fine-tuned?

The **last few convolutional/residual blocks** can be unfrozen and fine-tuned because they learn more task-specific features.

Use a **small learning rate** during fine-tuning.

Classification Head

Replace the original ResNet classification layer with:

Global Average Pooling

↓

Dense(128, ReLU)

↓

Dropout

↓

Dense(1, Sigmoid)

Training

Stage 1:

- Freeze ResNet layers.
- Train only the new classification head.

Stage 2:

- Unfreeze the last few ResNet layers.
- Fine-tune using a low learning rate.

Loss Function

For two classes:

Binary Cross Entropy

How does it improve performance?

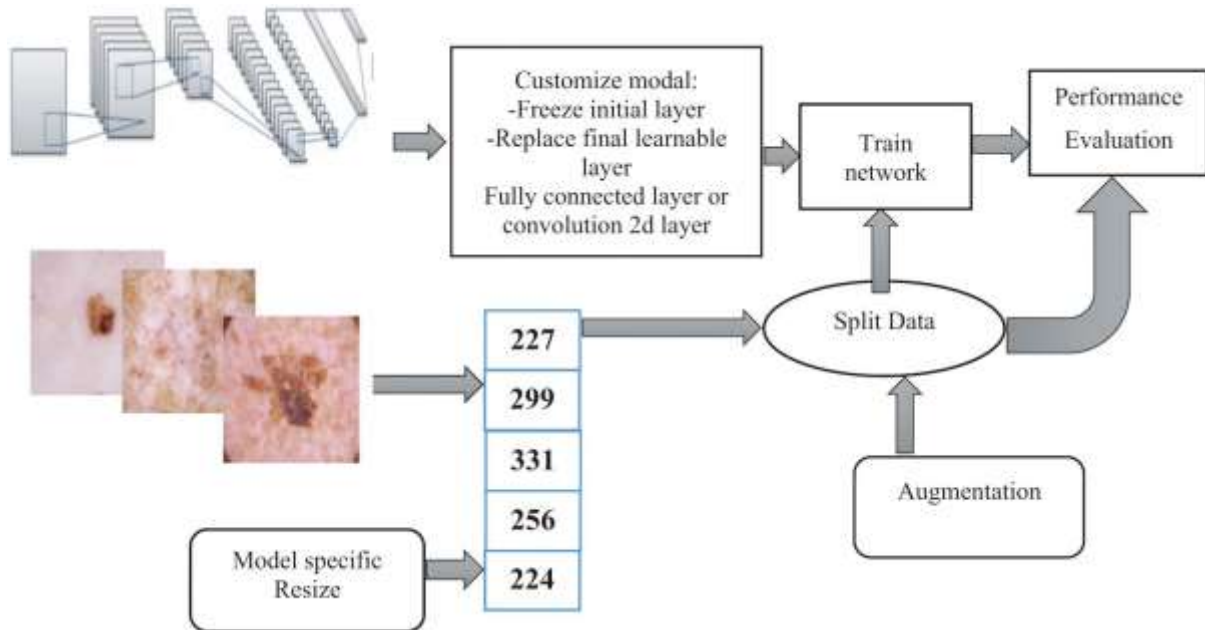
Transfer learning:

- Reduces training time.
- Requires fewer labeled images.
- Reduces overfitting.

- Reuses powerful pre-trained features.
- Fine-tuning adapts the model to skin-lesion characteristics.

Conclusion

A pre-trained ResNet50 with frozen early layers and fine-tuned deeper layers is an effective transfer learning strategy for skin cancer classification when labeled medical data is limited.



Q2. DenseNet and PixelNet for Road-Scene Segmentation

Scenario

The smart city system must identify:

- Roads
- Pedestrians
- Vehicles
- Traffic signs

at the **pixel level**.

This is a **semantic segmentation** problem.

Proposed Architecture

Road Scene Image

↓

Pre-processing

↓

DenseNet Feature Extractor

↓

Dense Blocks

↓

Transition Layers



PixelNet Segmentation Network

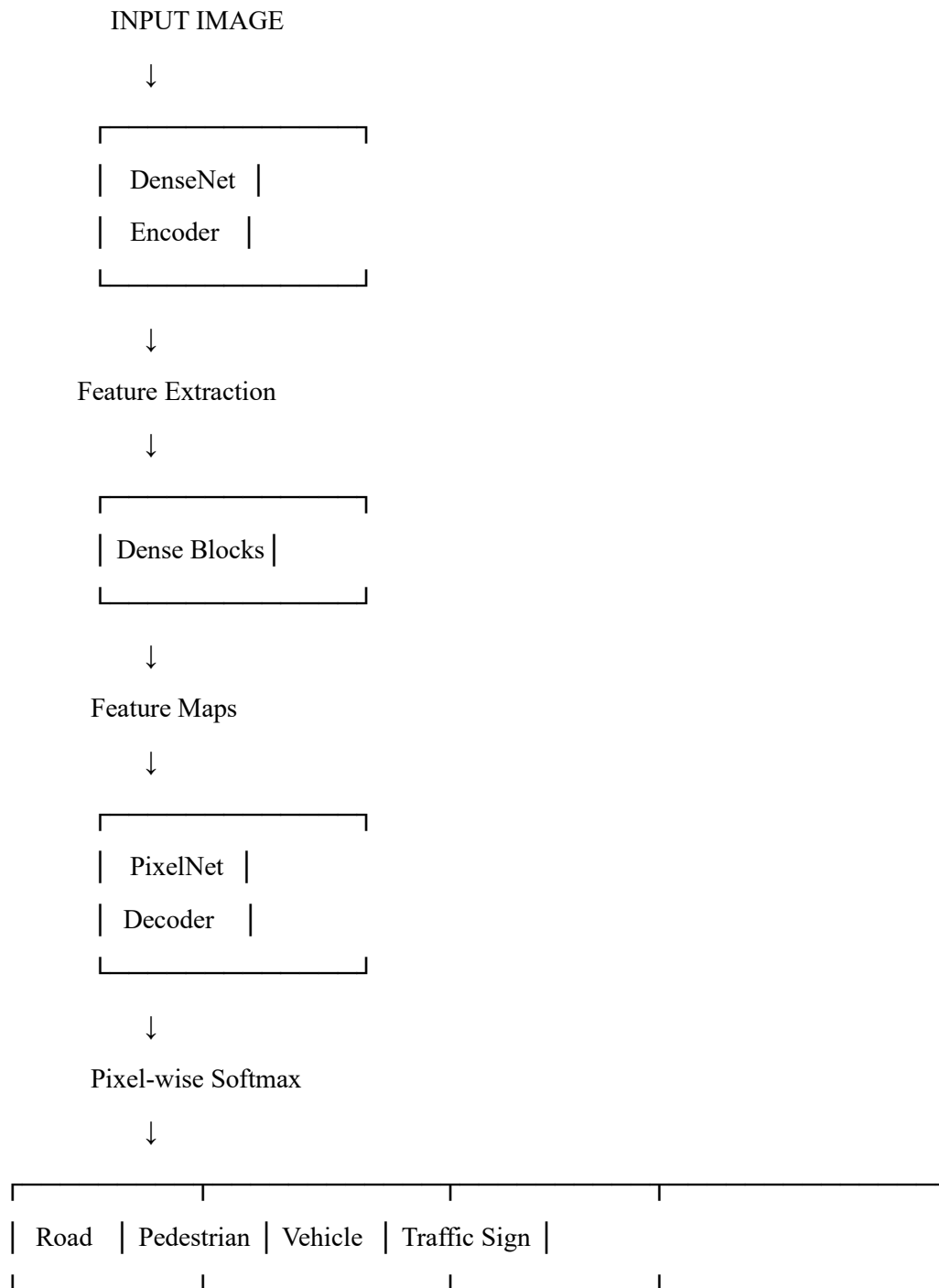


Pixel-wise Classification



Segmentation Map

Architecture Diagram



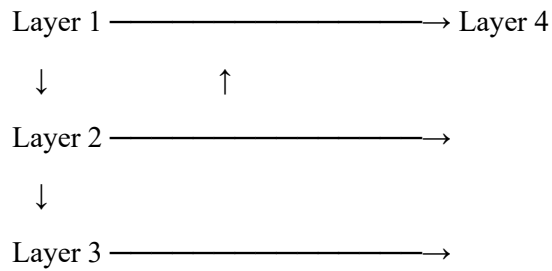
Role of DenseNet

DenseNet uses **dense connections**.

Instead of:

Layer 1 → Layer 2 → Layer 3 → Layer 4

DenseNet uses:



Each layer receives features from previous layers.

Advantages

1. Feature reuse

Early features such as edges and textures are reused by later layers.

2. Better gradient flow

Dense connections help gradients propagate through the network.

3. Fewer redundant features

The network does not need to relearn similar features.

Role of PixelNet

PixelNet performs **pixel-level classification**.

For every pixel, the model predicts a class.

Example:

Pixel → Road

Pixel → Road

Pixel → Vehicle

Pixel → Vehicle

Pixel → Pedestrian

Pixel → Road

Thus, instead of predicting only:

"There is a vehicle"

it predicts:

"These specific pixels belong to the vehicle."

How segmentation accuracy improves

DenseNet provides rich feature representations, while PixelNet performs pixel-wise prediction.

This helps distinguish:

- Small pedestrians
- Vehicles
- Traffic signs
- Road boundaries

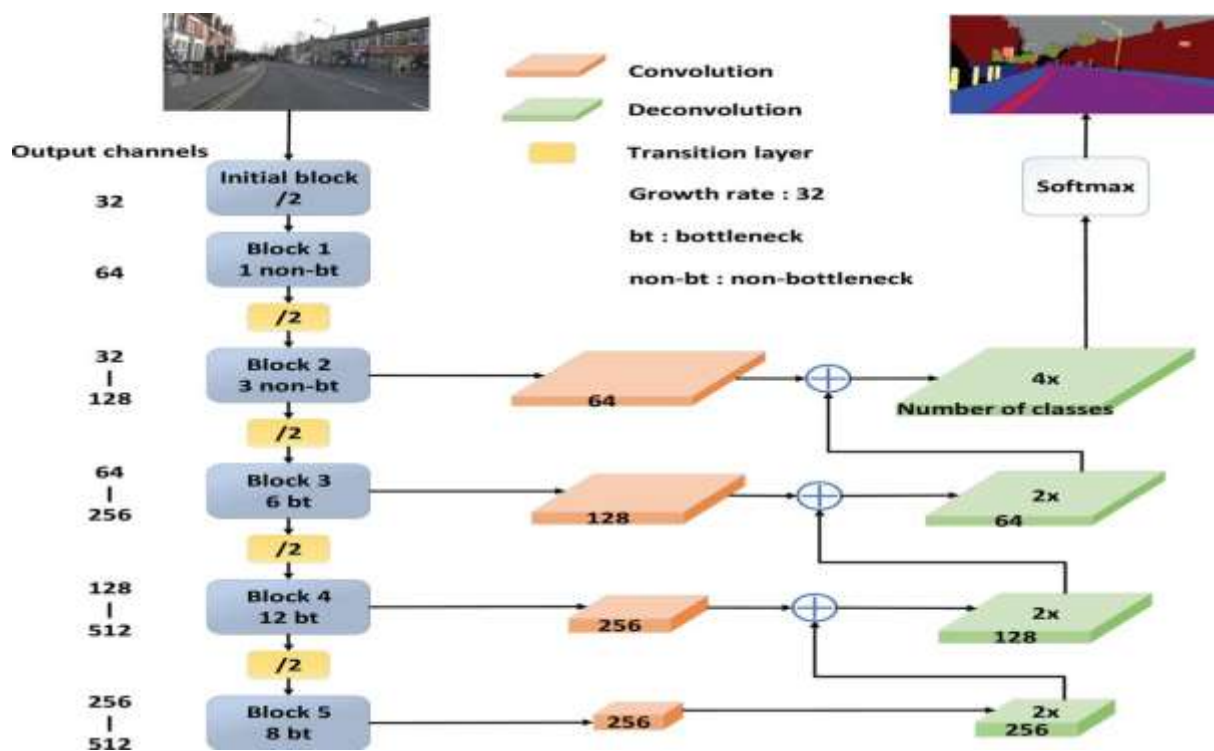
Computational Efficiency

DenseNet's feature reuse reduces redundant computation and parameters.

PixelNet can then use these extracted features for segmentation instead of learning all visual features again.

Conclusion

DenseNet acts as an efficient feature extractor through dense feature reuse, while PixelNet performs pixel-wise classification. Their combination provides accurate road-scene segmentation with efficient feature learning.



Q3. Bidirectional RNN + Encoder–Decoder for Multilingual Chatbot

Scenario

A multilingual chatbot must translate customer queries while maintaining contextual meaning.

A **Bidirectional RNN Encoder–Decoder** architecture is suitable.

Architecture

Customer Query

↓

Tokenization

↓

Word Embedding



Bidirectional RNN Encoder



Context / Hidden Representation



RNN/LSTM Decoder



Softmax Output



Translated Query

Detailed Architecture

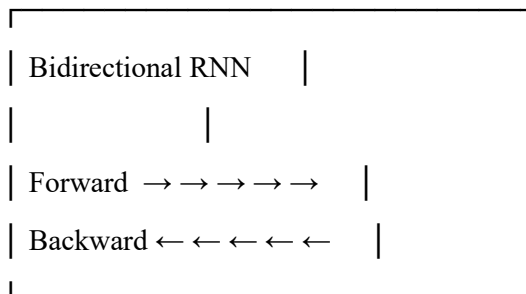
INPUT SENTENCE



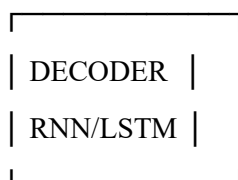
Tokenization



Embedding



Context Representation



Softmax Layer



Output Sentence / Translation

Encoder

The encoder reads the complete input sequence.

For example:

How → can → I → book → a → ticket?

It generates hidden representations containing information about the sentence.

Why Bidirectional RNN?

A normal RNN processes:

Past → Current

A Bidirectional RNN processes:

Forward: Word1 → Word2 → Word3 → Word4

Backward: Word4 → Word3 → Word2 → Word1

The two representations are combined.

Therefore, each word can use:

Previous context + Future context

Example

Consider:

"I saw a bank near the river."

The surrounding words help determine that **bank** refers to a riverside bank rather than a financial institution.

Decoder

The decoder generates the translated sentence one word at a time.

<START>

↓

Word 1

↓

Word 2

↓

Word 3

↓

...

↓

<END>

At each step, it uses:

- Previous output
- Decoder hidden state
- Encoder context

Output Layer

Softmax predicts the probability of the next word.

Word A → 0.10

Word B → 0.75 ← Selected

Word C → 0.15

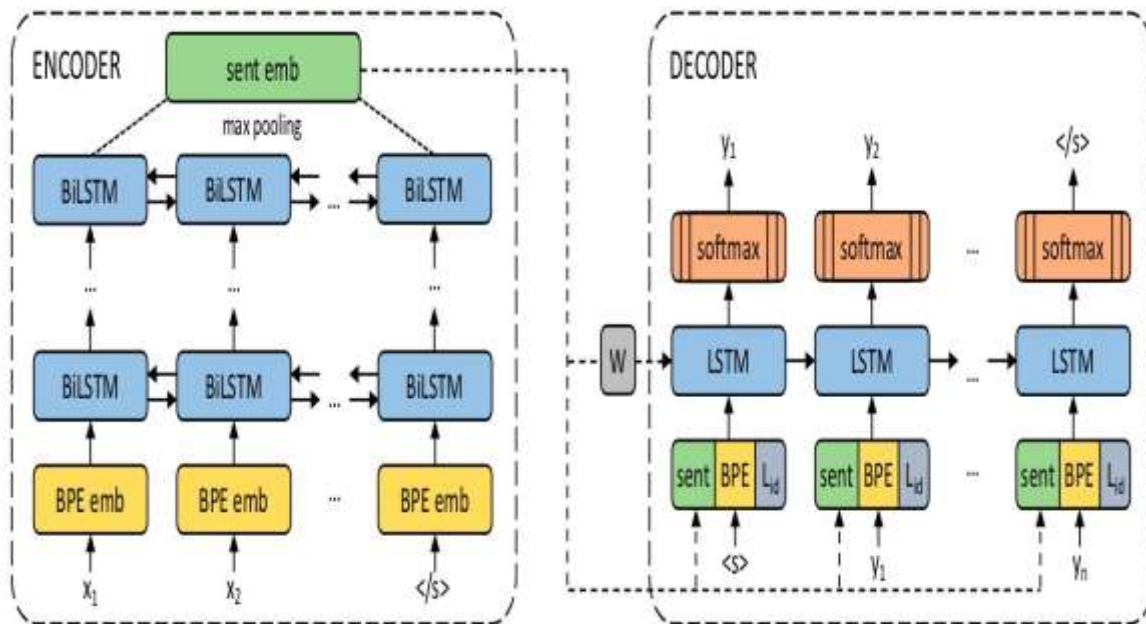
Why it improves translation?

Bidirectional processing:

- Captures past context.
- Captures future context.
- Understands complete sentence meaning better.
- Helps resolve ambiguous words.
- Improves contextual translation.

Conclusion

A Bidirectional RNN encoder captures information from both directions, while the decoder generates the target language sequentially. This improves contextual understanding and translation accuracy.



Q4. BPTT and LSTM for Stock Price Prediction

Scenario

Historical stock prices contain sequential information. The model needs to learn dependencies between previous and current market conditions.

Therefore, an **LSTM** is suitable.

Input Features

The model can use:

- Opening price
- Closing price
- High price
- Low price
- Trading volume

For example:

Day 1 $\rightarrow$ [Open, High, Low, Close, Volume]

Day 2 $\rightarrow$ [Open, High, Low, Close, Volume]

Day 3 $\rightarrow$ [Open, High, Low, Close, Volume]

...

Architecture

Historical Market Data

↓

Normalization



Sequence Formation



LSTM Layer



Dropout



Dense Layer



Output Layer



Predicted Stock Price

Example

Previous 30 Days



LSTM



Hidden Representation



Dense Layer



Next Day Price

What is BPTT?

BPTT = Backpropagation Through Time

It is the method used to train recurrent neural networks by extending backpropagation across the sequence's time steps.

Process

Suppose we have:

Day 1 → Day 2 → Day 3 → Day 4

The LSTM produces predictions at different time steps.

The error is calculated between:

Predicted value

vs

Actual value

Then the error is propagated backward through the time steps.

Forward:

Day1 → Day2 → Day3 → Day4

↓

Prediction

↓

Error

Backward:

Day4 ← Day3 ← Day2 ← Day1

The weights are updated to reduce the prediction error.

Why LSTM instead of Traditional RNN?

Traditional RNNs can suffer from:

Vanishing Gradient Problem

During BPTT, gradients may become extremely small when propagated through many time steps.

Therefore, traditional RNNs may have difficulty remembering long-term information.

LSTM Solution

LSTM contains:

- **Forget gate**
- **Input gate**
- **Output gate**
- **Cell state**

These mechanisms control what information should be:

Remembered

Forgotten

Updated

Output

Example

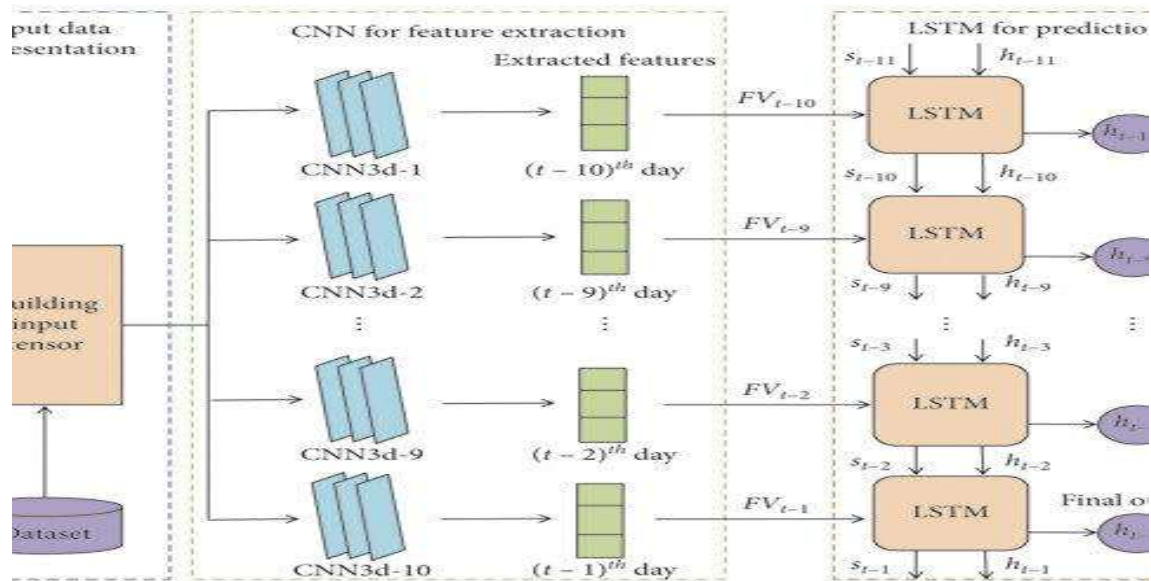
A stock price today may be influenced by:

Market behavior several days or weeks earlier.

LSTM can retain important historical information better than a traditional RNN.

Conclusion

LSTM is more suitable for stock prediction because its memory and gating mechanisms help preserve long-term dependencies and reduce the vanishing-gradient problem. BPTT trains the LSTM by propagating prediction errors through the sequence of time steps.



Q5. Integrated Deep Learning Solution for Automatic Video Captioning

Scenario

A media company wants to generate captions automatically for uploaded videos.

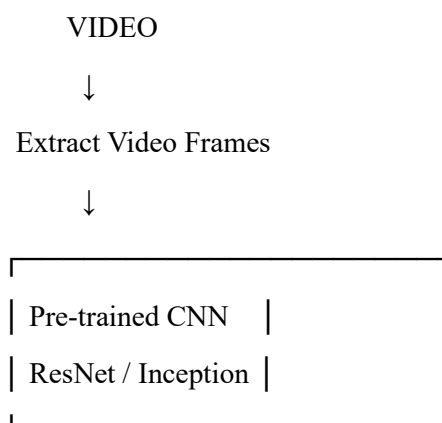
The system must understand:

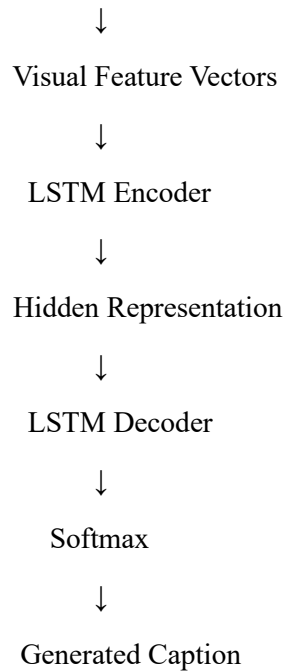
1. **What is happening visually?**
2. **How the visual events occur over time?**
3. **How to convert them into a sentence?**

Therefore, combine:

Transfer Learning + LSTM Encoder-Decoder

Overall Architecture





Example

Video:

A person is playing football.

The system generates:

"A person is playing football."

Component 1: Transfer Learning

Use a pre-trained CNN such as **ResNet**.

The CNN processes individual video frames.

Frame 1 → CNN → Feature Vector

Frame 2 → CNN → Feature Vector

Frame 3 → CNN → Feature Vector

Frame 4 → CNN → Feature Vector

Why transfer learning?

A pre-trained CNN already understands visual features such as:

- Edges
- Objects
- Shapes
- Textures
- Patterns

Therefore, the company does not need to train the visual feature extractor from scratch.

Component 2: LSTM Encoder

The visual feature vectors are provided to the LSTM encoder.

Frame 1 Feature



Frame 2 Feature



Frame 3 Feature



Frame 4 Feature



LSTM



Hidden Representation

The LSTM learns the **temporal information** in the video.

For example:

Person enters



Person picks ball



Person kicks ball



Person runs

LSTM understands the sequence of events.

Component 3: LSTM Decoder

The decoder converts the learned representation into natural language.

<START>



A



person



is



playing

↓

football

↓

<END>

At each step, the decoder predicts the next word.

Component 4: Softmax Output

The output layer calculates the probability of every word in the vocabulary.

Example:

playing → 0.65

running → 0.20

kicking → 0.15

The highest probability word is selected.

Role of Each Component

Component	Role
Video Frames	Provide visual information
Pre-trained CNN	Extract visual features
Transfer Learning	Reuse learned CNN features
LSTM Encoder	Understand temporal sequence
Hidden State	Store video context
LSTM Decoder	Generate caption word by word
Softmax	Predict next word

Why the Complete System Works

Transfer Learning

Provides powerful visual features without requiring huge amounts of video training data.

LSTM

Captures the temporal relationship between frames.

Encoder–Decoder

Converts the visual sequence into a natural-language sentence.

Complete Process

VIDEO



FRAMES



PRE-TRAINED CNN



VISUAL FEATURES



LSTM ENCODER



CONTEXT / HIDDEN STATE



LSTM DECODER



SOFTMAX



CAPTION

Conclusion

The integrated system uses transfer learning to efficiently extract visual features from video frames and an LSTM Encoder–Decoder to learn temporal relationships and generate captions word by word. This combination provides an effective solution for automatic video caption generation.

